

Stable commutator length and spectrum problems

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What is scl?

Algebraic definition. Let $w \in G$ be an element in a group.

$$\text{cl}_G(w) := \inf\{k \geq 0 \mid \exists a_i, b_i \in G, w = [a_1, b_1] \cdots [a_k, b_k]\} \in \mathbb{N}_{\geq 0} \cup \{\infty\}.$$

The **stable commutator length** of w is

$$\text{scl}_G(w) := \liminf_{n \rightarrow \infty} \frac{\text{cl}_G(w^n)}{n} \in [0, +\infty].$$

Remark. $\text{cl}_G(w) < +\infty \Leftrightarrow w \mapsto 0$ in $H_1(G; \mathbb{Z})$,

$\text{scl}_G(w) < +\infty \Leftrightarrow w \mapsto 0$ in $H_1(G; \mathbb{Q})$.

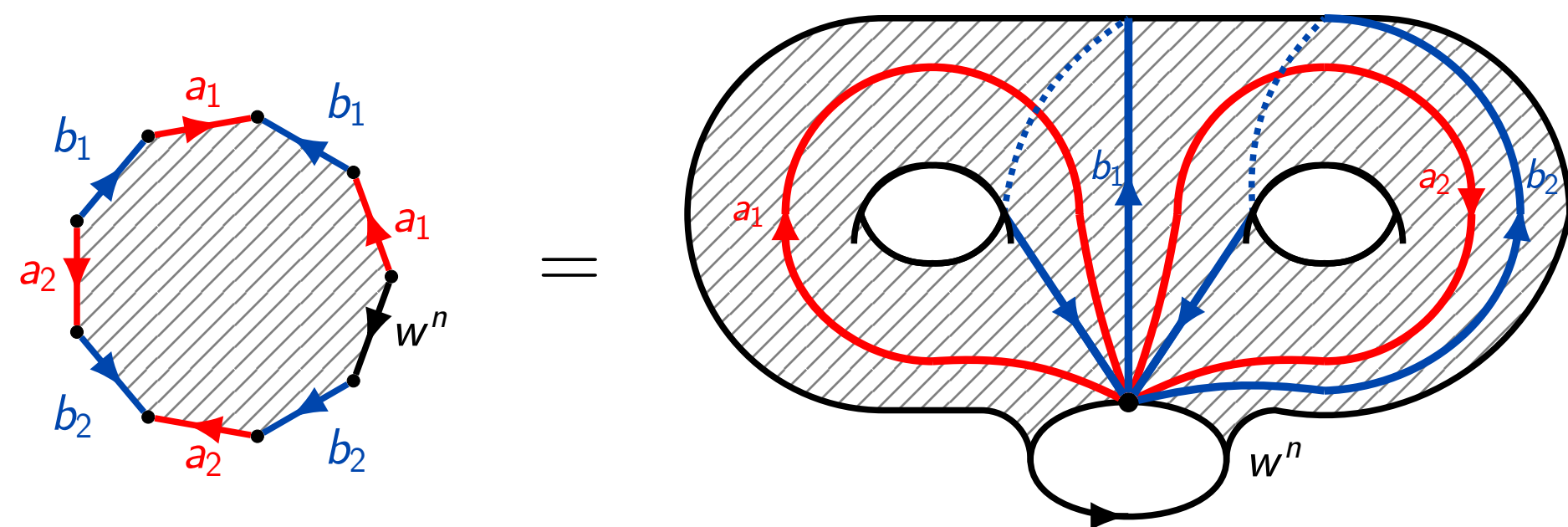
Topological interpretation. Let $G = \pi_1 X$, and $w = [\gamma : S^1 \rightarrow X]$. Then

$$\text{scl}_G(w) = \inf \left\{ \frac{-\chi^-(\Sigma)}{2 \deg(\partial\phi)} \mid \phi : (\Sigma, \partial\Sigma) \rightarrow (X, \gamma) \right\},$$

where $\phi : (\Sigma, \partial\Sigma) \rightarrow (X, \gamma)$ denotes the data of an oriented compact surface Σ , and continuous maps $\phi : \Sigma \rightarrow X$ and $\partial\phi : \partial\Sigma \rightarrow S^1$, such that

$$\begin{array}{ccc} \partial\Sigma & \xrightarrow{\quad} & \Sigma \\ \partial\phi \downarrow & \circlearrowleft & \downarrow \phi \\ S^1 & \xrightarrow{\quad \gamma \quad} & X \end{array} \quad \text{and} \quad \partial\phi_*[\partial\Sigma] = \underbrace{\deg(\partial\phi)}_{>0} [S^1] \text{ in } H_1(S^1; \mathbb{Z}).$$

Key picture. An equality $w^n = [a_1, b_1] \cdots [a_k, b_k]$ yields (here with $k = 2$):



Example 1. Given $a, b \in G$,

$$\text{scl}_G([a, b]) \leq -\frac{1}{2} \chi^- \left([a, b] \left(\begin{array}{c} \Sigma \\ \downarrow \phi \end{array} \right) \right) = \frac{1}{2}.$$

In fact, $\text{cl}_G([a, b]^n) \leq \lfloor n/2 \rfloor + 1$ (!)

Example 2. If $H = \langle a, b \mid [a, [a, b]] = [b, [a, b]] = 1 \rangle$, then

$$[a, b]^n = [a^n, b],$$

so $\text{scl}_H([a, b]) = 0$.

In fact, whenever G is **amenable**, we have $\text{scl}_G(w) = 0$ for all $w \in [G, G]$.

Quasimorphisms. A **quasimorphism** is a map $\phi : G \rightarrow \mathbb{R}$ such that

$$D(\phi) := \sup_{a, b \in G} |\phi(ab) - \phi(a) - \phi(b)| < \infty.$$

We say that ϕ is **homogeneous** if $\forall w \in G, \forall n \in \mathbb{Z}, \phi(w^n) = n \cdot \phi(w)$.

We denote $Q(G) := \{\text{homogeneous quasimorphisms } G \rightarrow \mathbb{R}\} \leq \mathbb{R}^G$.

Bavard Duality. $\text{scl}_G(w) = \sup \left\{ \frac{\phi(w)}{2D(\phi)} \mid \phi \in Q(G), D(\phi) \neq 0 \right\}.$

Spectral gaps

Definition. A group G is said to have an **ε -spectral gap** for scl if

$$\forall w \in G, \text{scl}_G(w) \in \{0\} \cup [\varepsilon, +\infty].$$

Moreover, G has a **strong ε -spectral gap** for scl if

$$\forall w \in G \setminus \{1\}, \text{scl}_G(w) \geq \varepsilon.$$

Theorem. Hyperbolic groups (Calegari–Fujiwara '10), MCGs (Bestvina–Bromberg–Fujiwara '16), 3-manifold groups (Chen–Heuer) have spectral gaps.

General philosophy. Negative curvature leads to large values of scl.

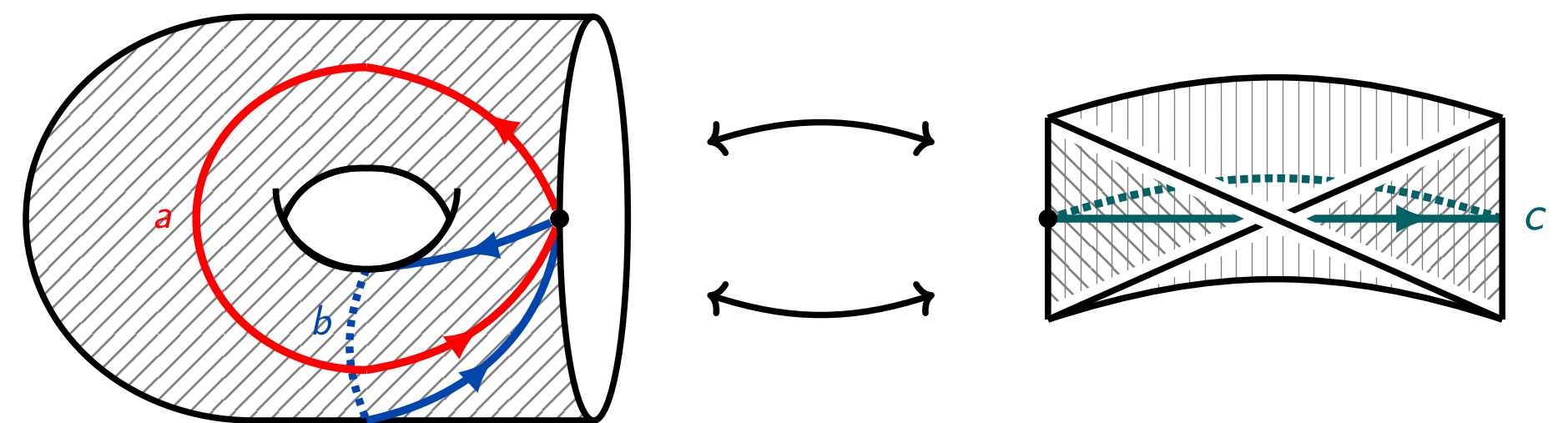
Strong spectral gap theorems (Duncan–Howie, Heuer, M.)

The following groups have a **strong $\frac{1}{2}$ -spectral gap**:

- Free groups (Duncan–Howie '91), and more generally
- Right-angled Artin groups (Heuer '19, M.).

Observation. Strong spectral gaps pass to subgroups. Therefore, a group G with an element w satisfying $0 < \text{scl}_G(w) < \frac{1}{2}$ cannot embed in a RAAG.

Example. Consider Dyck's surface $\Delta := (\mathbb{RP}^2)^{\#3}$:



We have $\pi_1 \Delta = \langle a, b, c \mid c^2 = [a, b] \rangle$, and $\text{scl}_{\pi_1 \Delta}(c) = \frac{1}{4}$. Therefore, $\pi_1 \Delta \not\hookrightarrow \text{RAAG}$.

Geometric method for spectral gaps (M.). Work with the topological interpretation of scl; fix $\gamma : S^1 \rightarrow X$ and start with $f : (\Sigma, \partial\Sigma) \rightarrow (X, \gamma)$.

- 1 Equip Σ with a certain geometric structure called an **angle structure**. This structure comes with a notion of **curvature**.
- 2 Estimate the curvature of Σ locally.
- 3 Sum up the curvatures and use a **Gauß–Bonnet formula** to deduce an estimate of $\chi(\Sigma)$.

Rationality

Rationality theorems (Calegari, Chen, ...)

Scl is **computable** and has **rational values** in

- Free groups (Calegari '09),
- Graphs of groups with infinite cyclic vertex and edge groups (Chen '20).

Question. What about closed surface groups?

Definition. Let $\alpha \in H_2(X, \gamma)$. The **relative Gromov seminorm** of α is

$$\|\alpha\|_1 := \inf \left\{ \frac{-2\chi^-(\Sigma)}{\deg(\phi)} \mid \phi : (\Sigma, \partial\Sigma) \rightarrow (X, \gamma) \right\},$$

where we now require $\phi_*[\Sigma, \partial\Sigma] = (\deg \phi)\alpha$ in $H_2(X, \gamma)$.

Proposition. We have

$$4 \text{scl}_G(w) = \inf \{ \|\alpha\|_1 \mid \alpha \in H_2(X, \gamma), \partial\alpha = [\gamma] \}.$$

In particular, if $X = S$ is a compact surface with $\partial S \neq \emptyset$, then $4 \text{scl}_{\pi_1 S}(w) = \|\alpha\|_1$ for $\alpha \in H_2(S, \gamma)$ the unique class with $\partial\alpha = [\gamma]$.

If however S is closed, there are many classes with $\partial\alpha = [\gamma]$.

Strategy for rationality in closed surface groups (M.). Let S be an o.c.c. surface, $\gamma : S^1 \rightarrow S$, and $w = [\gamma] \in \pi_1 S$.

- 1 Compute $\|\alpha\|_1$ for every $\alpha \in H_2(S, \gamma)$ with $\partial\alpha = [\gamma]$. This relies on the fact that $\|\cdot\|_1$ in (closed) surfaces often behaves like scl in free groups.
- 2 Minimise $\|\alpha\|_1$ over all α as above.

This will give $\text{scl}_{\pi_1 S}(w)$ by the above proposition.

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